

## **4-2: Project One**

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A wireframe is presented for the workflow of logging a watering interaction for gardens tracked by the app. User interaction elements are displayed and explained, as well as the flow from screen to screen as the water log task is performed. Rationale is provided for various design decisions. A discussion is given on how the application interface may be ported to Wear OS and to a touch-based kiosk.

### **Description and Explanation of Wireframe Screens**

The wireframes accompany this report as a separate attachment. It consists essentially of four different screens, A) the Home/Overview Screen, B) the Logging a Watering Event screen, C) the Specifying Which Crops were Watered screen, and D) Adding a Note screen. Note that seven screens are shown in the attachment, however three of these merely show the result of an animation or data update. The principles of Google's Material Design version 2 were substantially used. Navigation forward is done via Floating Action Buttons (FAB), chips, and buttons, as well as a "hamburger" main menu button. Navigation backward is done via the system-provided back button or gesture. The wireframes are in greyscale with no images (U.S. General Services Administration, 2013).

#### **Home/Overview screen**

The header is colored with brand primary color and white text. The top portion consists of a horizontal scroll panel of large activity buttons that represent various interactions with the garden that may be recorded and tracked. An overview portion below the action buttons summarizes how long it has been since several critical interactions have been performed. Each action summary may be clicked to display a detailed summary of history for that interaction. The colors of each summary is consistent with the color of the action button in the top.

### **Logging a Watering Event screen**

This screen begins the input workflow for recording the watering of crops in garden plots. The header, reworded after user comments during paper prototype interviews, informs the user the purpose of this screen. The current date and time are displayed as chip elements. Tapping either one brings up a date (or time) picker to allow recording a past watering. A Floating Action Button (FAB) is used to present the primary action (Google, n.d.-b) of saving the watering record. A secondary FAB is displayed above (in a more muted coloration) to add a note for the watering event. A list of all garden plots that the user has previously configured is displayed with a checkbox, signifying that this plot will be marked as watered when this record is saved. To minimize required user input for the most common cases (Apple, 2023b), default state is that all plots, and all crops contained, are watered. Under each plot, a chip element indicates which crops from the plot will be recorded as watered, for which the default is all crops. The chip for a plot can be tapped to change which crops were watered, which opens the *Specifying Watered Crops* screen for that plot. Any overflow in the list of plots is vertically scrollable. When the user taps the FAB to save the record, a system-specific snackbar (Google, n.d.-c) or toast message that states the record has been saved is temporarily shown.

### **Specifying Watered Crops screen**

To visually connect this form as supplementary to the Logging Water Event screen, an bottom sheet is used (Google, n.d.-a) which slides up from the bottom. It displays a header indicating the purpose of the sheet. Then it displays a list of all crops contained in the parent plot, each with a checkbox to indicate whether that crop was watered. The default is that all crops are watered, and a summary “select all” checkbox is available. Crop list overflow is vertically scrollable. A FAB for this sheet accepts the selections and closes the sheet, returning to the

overall *Logging a Watering Event* screen. To differentiate this FAB from the parent screen's FAB, it is colored with a different tint.

### **Adding a Note screen**

This screen is also implemented as a bottom sheet, presenting a instructional header and a single text multi-line text box labeled "Note". This is intended to be an unstructured area for any observations or things of note the user would like to record. Previous design iterations which included multiple note boxes for fertilizer, pests, etc proved to be confusing to users. Similar to the *Specifying Watered Crops* screen, and in the same coloration, a FAB is the primary action button which saves the note and closes the bottom sheet, leaving the user back at the *Logging a Watering Event* screen.

### **Adapting for a Digital Watch**

The small screen size, limited interaction controls, reduced battery life, and different user usage pattern cause challenges when adapting an existing app to a digital watch platform such as Apple Watch or Wear OS (Android). Only those functions from the parent app which can be performed simply and easily on the smaller watch face should be ported over (Apple, 2023a; Google, 2023). In the context of this app, the User Persona of Jamie is identified as the most likely to be interested in using a smart watch with this app, as she may have many more plots and might want to record items as she walks from plot to plot. A simple interface for Garden Checkup, in which she can use touch or the digital crown (Apple, 2023c) to cycle between garden plots, might be imagined. This would involve Jamie tapping the app to start a checkup session, then traveling to each plot one by one, recording items as she sees fit. Each plot screen might have a list of icons (watering, harvesting, fertilizing, pests, etc) each with a date indicating the last time this interaction was performed for the plot. Tapping, for example, the pests icon,

would record the fact that something relating to pests was notable for this plot. Due to the small screen of the device, it might not be feasible to implement a textual input to record unstructured notes, but these might be added later from the app on the larger mobile device. Additionally, implementing notifications if some number of days have elapsed since the last time an important garden interaction has been performed, such as watering or fertilizing would add value and be glanceable.

### **Adapting for a Touch-Based Kiosk**

A large-scale touch-based screen would likely be a more natural fit for this app than a watch face. The larger size and (often) more powerful performance specifications of this class of devices allows greater flexibility and a wider range of options of user interface layouts and elements. The garden checkup workflow could be made much more efficient by utilizing the *supporting pane* canonical layout (Google, 2024b). This consists of a large primary content pane alongside a smaller supporting pane. The supporting pane would list and allow selection of each plot. The primary content pane would present controls allowing recording of any of the interactions with this plot, including watering, fertilizing, planting, harvesting. There would also be more space to allow entering notes. A significant improvement with this design would be incorporating additional tools in the supporting pane, including the ability to quickly review the history of previous interactions with this specific plot. Following convention, navigation to other tools in the app would be through a slide-out navigation drawer (Google, 2024a). The layout would be responsive to orientation, with the supporting pane being above the primary if in portrait, and to the right if in landscape. This use case, and the ability to more easily collect and input many kinds of interactions from one screen, would also suggest this as being more likely to be used by Jamie for tracking her larger gardens in a structured and professional manner.

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